

Measuring Consumer Preferences for Used Battery Electric Vehicles

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1 Motivation and Research Question

Used vehicles account for approximately 70% of all vehicle purchases in the United States [1], yet the used electric vehicle (EV) market has received comparatively little attention from researchers and policymakers. Meanwhile, used EV sales rose 34.2% year-over-year as of February 2025, and average days' supply fell 21.5%, signaling strong and accelerating demand [2]. A healthy used EV market matters across multiple dimensions: it extends the useful life of vehicles and batteries, supports a circular economy, and lowers the financial barrier for middle and lower-income households to enter the EV market [3].

Despite this momentum, consumer hesitation persists. Survey data indicate that 52% of U.S. adults are unlikely to consider buying a used EV, compared to only 28% who are open to the idea [4]. Beyond the range anxiety and charging infrastructure concerns shared with new EVs, the used market carries additional risks: uncertainty about vehicle history and condition, and perceptions of lower reliability. Central to these concerns is the battery — the most expensive and least observable component of a used battery electric vehicle (BEV). Unlike mileage or vehicle age, which serve as reasonable proxies for wear in conventional vehicles, battery degradation varies substantially across EVs of the same model year and mileage, creating a significant information asymmetry at the point of sale [5].

This study addresses the following research questions:

1. How much are consumers willing to pay (WTP) for a used BEV powertrain relative to a comparable conventional vehicle?
 2. How does WTP vary with key vehicle attributes — driving range, vehicle age, and price budget?
 3. How do preference structures differ between likely and unlikely EV adopters?
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2 Background and Contribution

Most EV preference research focuses on new vehicle buyers and treats price sensitivity as uniform. The used-vehicle market has its own buyer population, its own price dynamics, and its own barriers to EV adoption. Affordability, vehicle condition, and reliability dominate used-vehicle purchase criteria [6], and average used-vehicle prices reached \$25,547 in April 2025 [7], putting many segments under acute cost pressure.

Prior work on used EV adoption identifies three broadly distinct buyer segments: those for whom purchase cost is a binding constraint, those apprehensive about EV technology, and those drawn by a favorable cost-benefit ratio relative to new EV depreciation [8]. Demographic profiles of current used EV owners closely mirror new EV owners — disproportionately male, middle-aged, highly educated, and higher-income — suggesting the used market has not yet broadly reached lower-income buyers [9]. Battery-related concerns (degradation, state-of-health, replacement cost) and charging access are the dominant pre-purchase barriers for lower-income segments in particular [10].

This study contributes the first quantitative estimates of WTP for powertrain type in the used BEV market for U.S. disaggregated by budget, adoption propensity, vehicle type (car vs. SUV), driving range, and vehicle age. The discrete choice experiment (DCE) design allows us to recover implicit prices for attributes in the used-vehicle context.

3 Data and Methods

Survey design. The survey targets U.S. residents aged 18 or older who intend to purchase a used car or SUV within the next two years. The core instrument is a DCE in which respondents complete six repeated choice tasks, each presenting three used vehicle alternatives (plus a “none of the above” option) with varying attributes. Attributes include vehicle mileage, powertrain (conventional ICEV, gas hybrid HEV, or battery electric BEV), operating cost (cents per mile), model year, and purchase price. Price levels are segmented by vehicle type and budget tier: low-budget cars (\$10,000–\$20,000), high-budget cars (\$20,000–\$40,000), low-budget SUVs (\$15,000–\$25,000), and high-budget SUVs (\$25,000–\$45,000).

Beyond the DCE, the survey collects current vehicle ownership details, future purchase intentions, EV knowledge and attitudes (5-point Likert scales covering range anxiety, environmental beliefs, cost perceptions, and battery concerns), and standard demographic variables (income, housing type, education, employment status).

Data collection and sample. Data were collected online between December 2025 and April 2026 across two research platforms. After removing responses that failed attention checks, showed within-survey inconsistencies, or contained uninformative open-ended text, the final

analytic sample is **2,916 respondents**. Median survey completion time was approximately 14 minutes. Respondents are distributed across the United States.

Estimation. We estimate a multinomial logit (MNL) model via maximum likelihood, treating each choice task as a utility-maximizing decision over observed vehicle attributes. Willingness to pay for each attribute is derived as the ratio of its coefficient to the price coefficient, representing the dollar amount a respondent would pay or require as compensation for a one-unit change in that attribute. WTP is estimated separately across segments defined by budget tier, adoption propensity, vehicle type, and vehicle age.

4 Results

Note: This study is a work in progress. The results presented below reflect initial findings from the MNL model. Full segmentation analysis and treatment effect estimates are ongoing.

4.1 WTP for used BEVs by adoption preference, vehicle type, and range

Figure 1 presents modelled WTP for used BEVs at a fixed vehicle age of 2 years, disaggregated by adoption preference (likely vs. unlikely EV adopter), vehicle type (car vs. SUV), and driving range (100, 200, and 300 miles). The red dashed line at \$0 marks the threshold above which consumers value a BEV at a premium relative to a comparable conventional vehicle; below \$0 they require a discount.

Modelled WTP for EVs by Adoption Preference, Vehicle Type, and Range (Vehicle Age is 2 years)



Figure 1: Modelled WTP for used BEVs by adoption preference, vehicle type, and range (vehicle age = 2 years). Points show point estimates; horizontal lines show 95% confidence intervals.

Likely adopters show substantially higher WTP than unlikely adopters across all range and vehicle-type combinations. At 300 miles, likely car adopters are the only group whose WTP crosses into positive territory (approximately +\$10,000 to +\$15,000), meaning they would pay a premium for a used BEV over an equivalent conventional car. Unlikely adopters at 100-mile range — the lowest-range, lowest-appeal combination — require discounts approaching -\$35,000 to -\$40,000. SUVs carry consistently lower WTP than cars within each segment.

4.2 WTP by adoption preference, vehicle type, and age (range fixed at 300 miles)

Figure 2 holds range constant at 300 miles and varies adoption preference, vehicle type, and vehicle age (2, 5, and 8 years).

Modelled WTP for EVs by Adoption Preference, Vehicle Type, and Age
(Vehicle Range is 300 miles)

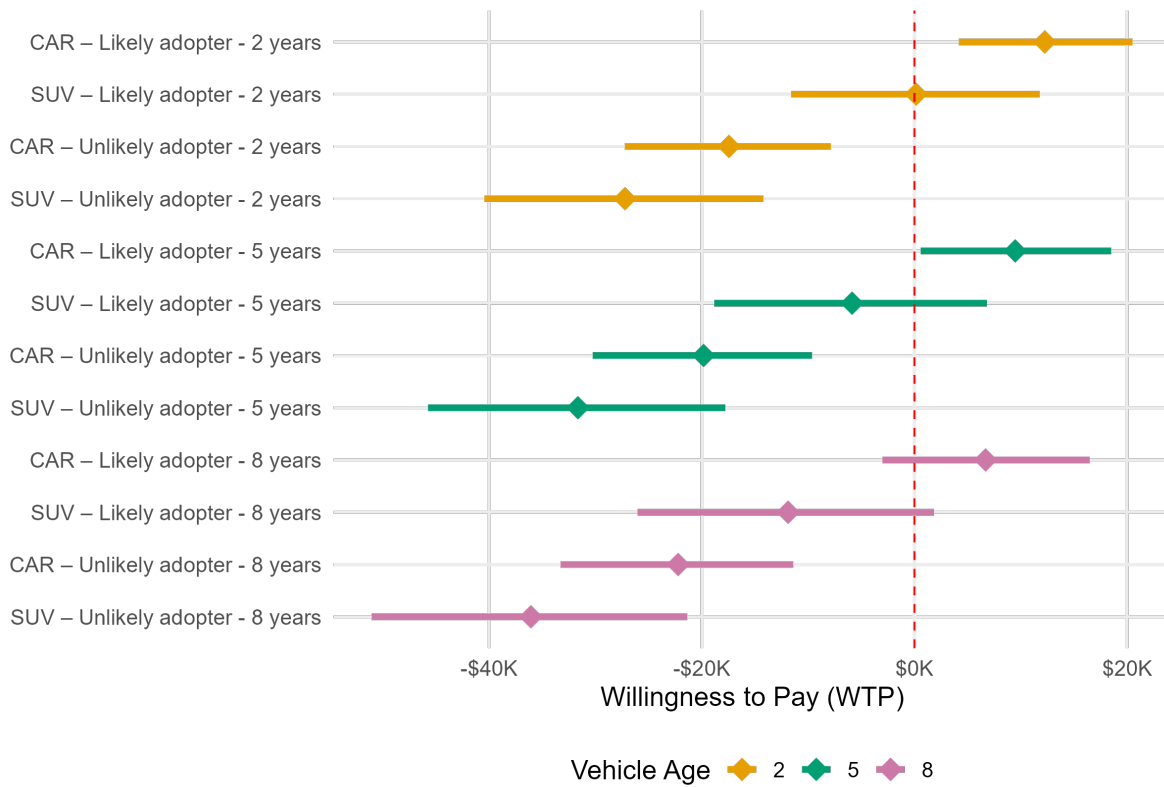


Figure 2: Modelled WTP for used BEVs by adoption preference, vehicle type, and vehicle age (range = 300 miles).

This plot shows the widest spread of any segment combination. Vehicle age exerts a large and monotonic effect on WTP: each additional cohort of vehicle age (2 → 5 → 8 years) moves WTP meaningfully further into negative territory. The age penalty is especially pronounced among unlikely adopters, whose WTP drops sharply across age groups. Likely adopters are more resilient to aging: their WTP remains near or above \$0 even at 5 years old for cars. For 8-year-old SUVs among unlikely adopters, required discounts approach -\$40,000.

4.3 WTP by budget, vehicle type, and age (range fixed at 300 miles)

Figure 3 focuses on the high-budget segment at 300-mile range and shows how WTP varies across vehicle type and age.

Modelled WTP for EVs by Budget Preference, Vehicle Type, and Age
(Vehicle Range is 300 miles)

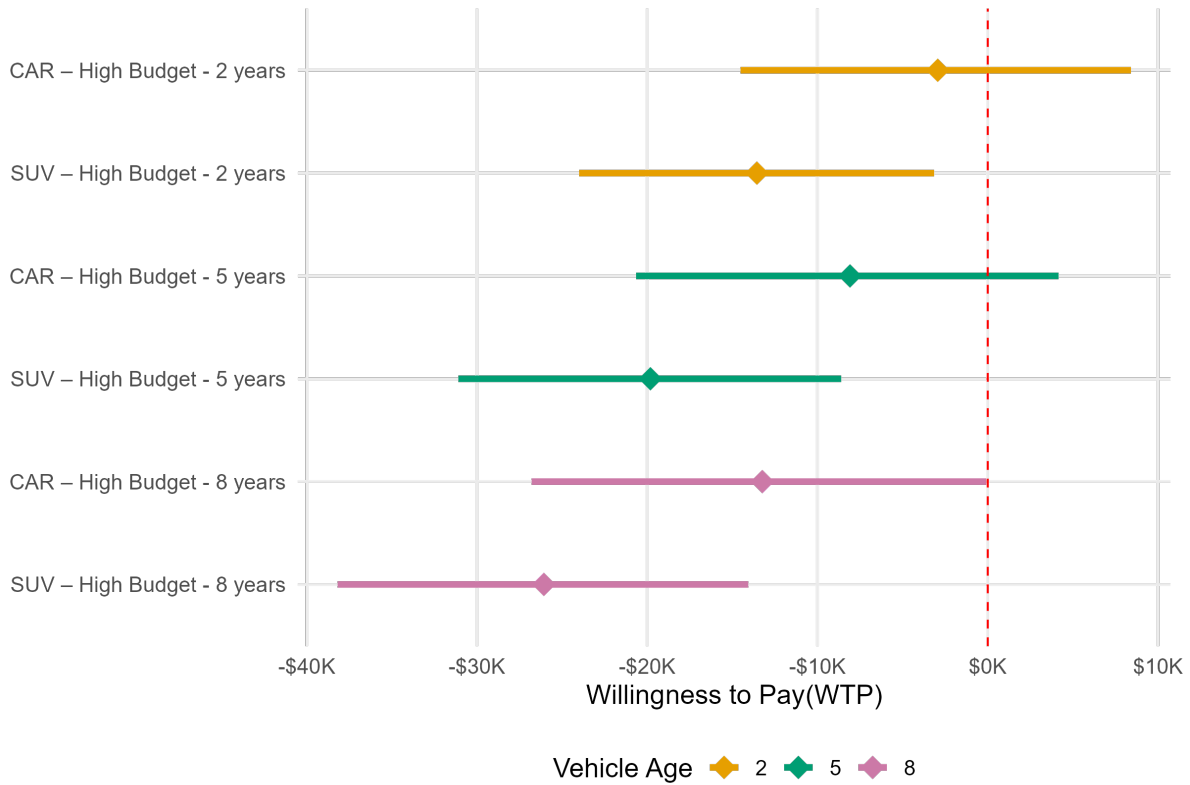


Figure 3: Modelled WTP for used BEVs by vehicle type and age — high-budget consumers, range fixed at 300 miles.

High-budget car buyers at 2 years old have WTP close to \$0 with a wide confidence interval that crosses into positive territory, suggesting this segment can potentially support near-parity pricing for used BEVs at long range. The age penalty remains large: moving from 2 to 8 years shifts WTP by approximately -\$15,000 to -\$20,000 within the high-budget segment. SUVs lag behind cars at every age level, and confidence intervals are notably wide throughout this segment.

4.4 Summary of initial findings

Across all model specifications estimated to date, three attributes dominate WTP:

Factor	Direction of effect
Driving range (100 → 300 miles)	Substantially increases WTP
Vehicle age (2 → 8 years)	Substantially decreases WTP
Adoption preference (unlikely → likely)	Substantially increases WTP
Budget (low → high)	Moderately increases WTP
Vehicle type (SUV vs. car)	SUV WTP is consistently lower

The only consumer segment that currently shows positive WTP — willingness to pay a premium for a used BEV over an equivalent conventional vehicle — is likely adopters choosing 300-mile cars at 2 years of age. For most other segments, used BEVs must be priced at a discount relative to conventional alternatives to attract buyers. These findings underscore the importance of range, vehicle age, and consumer adoption propensity as the primary levers for the used BEV market.

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